

Chapter - 8
More on Python (Part-1)
Exercise Question answers

1. Explain the different programming environments available for python and their unique features.

Ans:- The following are the different programming environments for Python.

- IDLE's Interactive Shell: Allows interaction with the operating system using the command window.
- IDLE's Editor: Helps in creating and editing Python source files using the built-in editor.
- Notepad: A simple text editor for writing Python programs.
- VS Code (Visual Studio Code): A free, open-source text editor available for multiple platforms.
- JupyterLab: A web-based, interactive development environment for notebooks, code and data.
- Google Colab: A cloud-based tool from Google that allows writing and executing Python code without installation.

2. Discuss the types of statements in python and provide examples of single-line and multi-line statements.

Ans:- Statements in Python are instructions written in source code for execution. Following are the types of Statements in Python.

- Single-line statement: An instruction that executes entire code on one single line of code.

```
print("Hello, Python!")
```

- Multiple statements using a semicolon (;) : Two or more independent instructions written on the exact same line.

```
print("I have been learning"); print("Python"); print("for 10 days")
```

- Multi-line statements : A single instruction that breaks and spans across two or more lines.

```
print("I have been learning\n\
Python\n\
for 10 days")
```

Note: \n is a newline escape sequence and \ at the end of a line allows continuation on the next line.

3. What are tokens in python? Describe the various types of tokens with suitable examples.

Ans:- Tokens are the smallest program elements that the compiler recognises. Following are the various types of tokens.

- Keywords: Reserved words in Python, e.g., if, else, while, def.
- Identifiers: Variable names like name, age.
- Literals: Fixed values such as 100, "Hello", True.

- Punctuators: Symbols like , ; { } [].
- Operators: Used for operations, e.g., + - * / == !=.

Example

```
age = 25 # Identifier  
if age > 18: # Keyword and Operator  
    print("Adult") # Literal
```

4. What are literals in python ?Explain different types of literals with examples.

Ans:- Literals are constant values assigned to variables. Following are the different types of literals.

- Integer Literals: Whole numbers like 10, -5, 1000.
- Floating-Point Literals: Decimal numbers like 3.14, -0.001.
- Character Literals: A single letter or symbol, e.g., 'A', "b".
- String Literals: Text enclosed in quotes, e.g., "Hello", 'Python'.
- Boolean Literals: True or False.
- None Literal: Represents an absence of value (None).

Example

```
marks = 95 # Integer Literal
pi = 3.14 # Floating-point Literal
name = "John" # String Literal
is_passed = True # Boolean Literal
result = None # None Literal
```

5. Describe the role of escape sequences in Python programming and provide examples of their usage.

Ans:- Escape sequences allow special formatting in text output.

Escape Sequence	Description
\n	Newline (Moves text to the next line)
\t	Tab (Adds a horizontal space)
\b	Backspace (Deletes previous character)
\'	Single quote (Allows using ' in a string)
\"	Double quote (Allows using " in a string)
\\	Backslash (Prints \ in a string)

Example

```
print("Hello\nWorld") # Outputs:
# Hello
# World
```

```
print("Python\tProgramming") # Outputs: Python  Programming
```

The following are the differences between implicit and explicit line continuation.

6. Explain the difference between implicit and explicit line continuation in Python with examples.

Ans:- Python allows implicit and explicit ways to continue long statements.

Feature	Implicit Line Continuation	Explicit Line Continuation
Definition	Uses parentheses (), brackets [] or braces {} to break statements.	Uses a backslash \ to indicate continuation.
Usage	Common in function calls, lists and dictionaries.	Used when implicit continuation is not applicable.
Example	sum = (10 + 20 + 30 + 40)	sum = 10 + 20 + \ 30 + 40